



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to TPN COMPANY Indian Oil Dealers

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

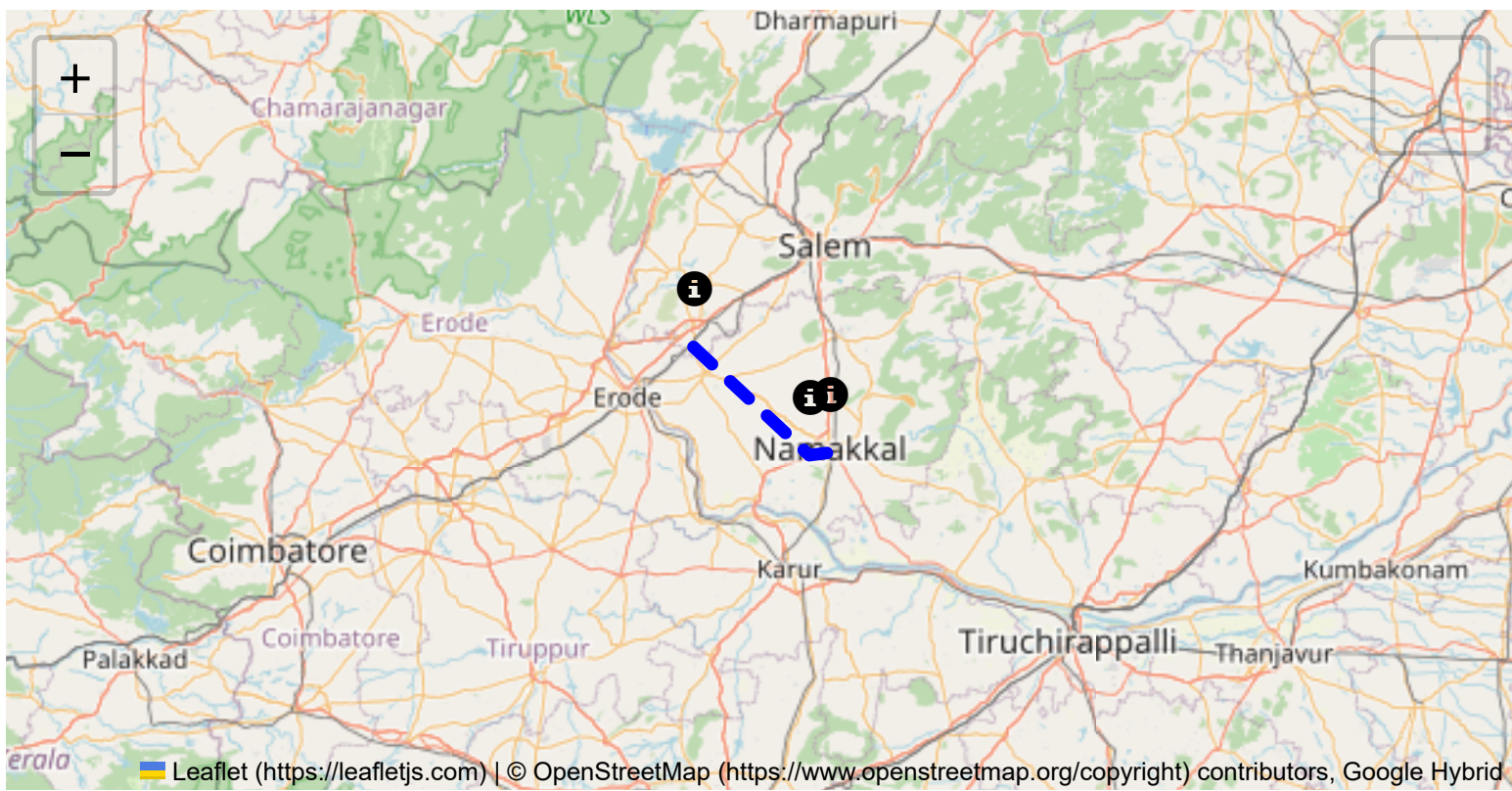
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

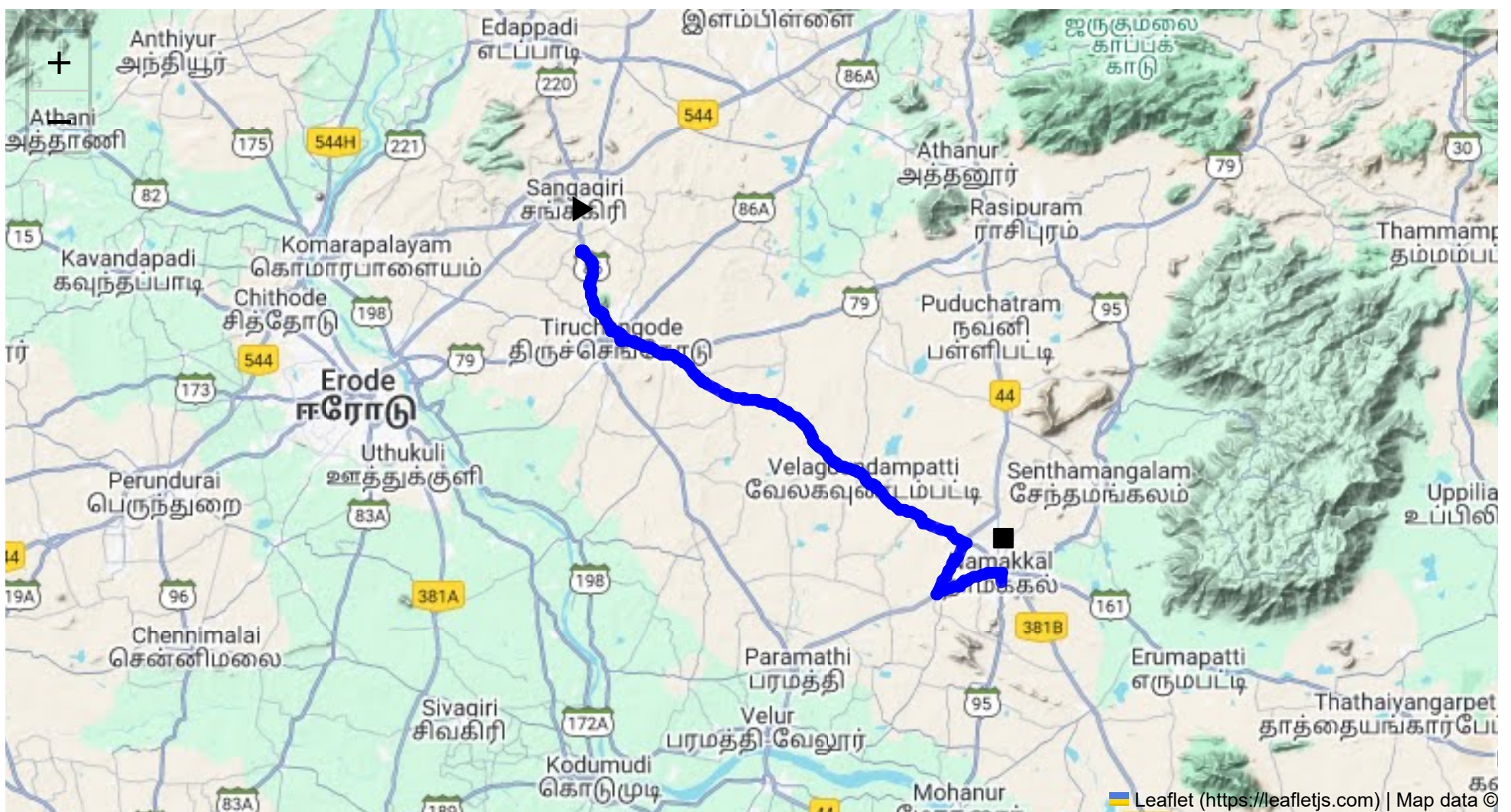
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 51.52 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.211609, 78.167714)



Welcome to the Journey Risk Management Study

Route Overview:

The route from CVQF+23W, Sangagiri to 6569+J34, K K Nagar in Namakkal via 643F+XH2, Vallipuram covers approximately 51.52 kilometers. It passes through state highways and rural roads typical of the Tamil Nadu region, combining developed and semi-rural areas.

Weather Conditions:

Typical weather conditions in this region may vary by season. During the monsoon (June to September), heavy rainfall can lead to slippery roads and occasional flooding, which could pose risks for heavy vehicles. The remainder of the year, the weather is generally dry but can be hot, particularly from March to May.

Traffic Patterns:

- **Peak Hours:** Mornings (7:30-9:00 AM) and evenings (5:00-7:30 PM) typically experience increased traffic.
- **Congestion-Prone Areas:** Sangagiri town center and entry into Namakkal might experience congestion. Unexpected delays can also occur at toll booths or check-posts.

Road Quality and Infrastructure:

- **Sangagiri to Vallipuram:** Generally good road conditions, with some narrow stretches.
- **Vallipuram to Namakkal:** Occasionally, the road might have potholes or uneven surfaces. Regular updates on road work are necessary.

Suggestions for Alternative Routes:

In case of emergencies or extreme traffic, consider using NH 44 for faster access, keeping in mind the potential for higher traffic densities, but better infrastructure.

Local Regulations on Hazardous Material Transport:

Tamil Nadu imposes restrictions on the transport of hazardous materials in urban areas during peak hours. Compliance with permits and documentation is essential. Specific routes might be designated for hazardous materials.

Historical Incidents:

While specific data for this precise route may not be widely reported, incidents involving heavy vehicles typically include accidents due to road conditions and narrow roads, emphasizing the need for careful handling.

Environmental Considerations:

- **Sensitive Areas:** Passing through rural and semi-urban landscapes, the route may impact local farmlands and habitats. Care should be taken to avoid potential spills or leaks of hazardous materials.

Communication Coverage:

- **Potential Dead Zones:** Rural stretches between Vallipuram and Namakkal may experience weak mobile network coverage, affecting communication.

Estimated Emergency Response Times:

- **Sangagiri to Vallipuram:** Approximately 20-30 minutes for response teams during clear conditions.
- **Vallipuram to Namakkal:** Response time could be longer, around 30-45 minutes, due to rural settings and road quality.

Overall Risk Assessment:

The route presents moderate risks primarily due to weather conditions, road quality, and potential traffic congestion. Adherence to local regulations and awareness of alternative routes are crucial for reducing risks. Prior planning, especially during monsoon seasons and peak traffic hours, can mitigate potential hazards. Regular monitoring of traffic updates and weather conditions should be part of the travel routine.

Frequent traveler updates can further enhance safety, while local knowledge of short routes or detours can be invaluable. Training for emergency responses and hazard management should be prioritized to effectively manage any incidents involving hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	11.20387, 78.12203	15 KM/Hr
2	Turn	High	11.43811, 77.87348	15 KM/Hr
3	Turn	Medium	11.43956, 77.87340	30 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
6	Turn	Medium	11.38342, 77.89975	30 KM/Hr
7	Turn	High	11.38186, 77.90258	15 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Blind Spot	Blind Spot	11.23936, 78.14267	10 KM/Hr
10	Turn	High	11.22817, 78.13576	15 KM/Hr
11	Turn	High	11.22822, 78.13561	15 KM/Hr
12	Turn	Medium	11.20882, 78.12464	30 KM/Hr
13	Turn	Medium	11.21882, 78.16760	30 KM/Hr
14	Turn	Medium	11.21926, 78.16775	30 KM/Hr
15	Blind Spot	Blind Spot	11.21933, 78.16773	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
16	Turn	High	11.21935, 78.16804	15 KM/Hr
17	Blind Spot	Blind Spot	11.21912, 78.16824	10 KM/Hr
18	Turn	High	11.21889, 78.16777	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
15	clinic	Pranauv Clinic	11.215361, 78.1484232	30 km/h	Medium
16	hospital	Vairam Hospital,Namakkal	11.217603, 78.153269	30 km/h	Medium
17	hospital	Maharishi Eye Hospital	11.219304, 78.1610484	30 km/h	Medium
18	clinic	Rithiksha Physio Care	11.2189918, 78.159957	30 km/h	Medium
19	clinic	Sri Sara ENT Clinic	11.218949, 78.159922	30 km/h	Medium
20	hospital	Namakkal Neurocare Hospital	11.218663, 78.15976	30 km/h	Medium
21	hospital	Vasantham Hospital	11.2187521, 78.161599	30 km/h	Medium
22	hospital	Shanmuga Heart Centre	11.22124, 78.163711	30 km/h	Medium
24	clinic	Krishnan Urology Clinic	11.2184523, 78.1643654	30 km/h	Medium
25	hospital	Natasan Hospital	11.2202016, 78.1639129	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
26	hospital	Children's Hospital	11.220084, 78.1639673	30 km/h	Medium
27	hospital	Nirmala Hospital	11.2179361, 78.1665001	30 km/h	Medium
28	hospital	Surya Hospital, Namakkal	11.2178506, 78.1660375	30 km/h	Medium
31	hospital	Shanthi Hospitals, Namakkal	11.2190661, 78.1685735	30 km/h	Medium
32	hospital	Government Hospital	11.2168924, 78.169068	30 km/h	Medium
33	police	Police Station	11.2185353, 78.1695347	30 km/h	Medium
34	police	Women Police Station	11.2183133, 78.1692677	30 km/h	Medium
35	hospital	Ganga Hospital	11.21742, 78.170272	30 km/h	Medium
36	hospital	SMK Hospital	11.2172759, 78.1706866	30 km/h	Medium
37	hospital	Poornima Hospital	11.215533, 78.16911	30 km/h	Medium
38	clinic	ENT Clinic	11.215786, 78.170545	30 km/h	Medium
39	hospital	Chellam Hospitals	11.2153614, 78.1684456	30 km/h	Medium
40	hospital	Deepthi Hospital	11.2152757, 78.1680221	30 km/h	Medium
41	hospital	Sabari Childrens Hospital	11.2151465, 78.1680375	30 km/h	Medium
42	hospital	Sri Ragavendra Hospital	11.2151734, 78.1703431	30 km/h	Medium
44	hospital	R.K. Hospital, Namakkal	11.2143252, 78.1678846	30 km/h	Medium
45	hospital	Cm Hospital	11.2136963, 78.1680022	30 km/h	Medium
46	hospital	C.M. Speciality Hospital and Jaya Lizarov Centre	11.2122571, 78.1697107	30 km/h	Medium
47	hospital	Maharaja Speciality Hospital	11.210755, 78.166997	30 km/h	Medium
48	hospital	Kiruba Hospital	11.2107891, 78.1677358	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
23	school	Anna Nehru Matriculation School	11.216056, 78.164447	30 km/h	Medium
29	school	Government Girls Higher Secondary School	11.2179186, 78.1672383	30 km/h	Medium
30	school	Namakkal Government Boys Higher Secondary School	11.2150472, 78.1669273	30 km/h	Medium
43	school	Kandasamy Kandar Matriculation and Higher Secondary School	11.2137988, 78.1698234	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.20387, 78.12203



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38342, 77.89975



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.23936, 78.14267



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.22817, 78.13576



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.22822, 78.13561



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.20882, 78.12464



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.21882, 78.16760



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.21926, 78.16775



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.21933, 78.16773



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.21935, 78.16804



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.21912, 78.16824



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.21889, 78.16777
